PhyGRU Application: Vehicle Dynamics Modelling

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1 Introduction

This work investigates the application of the PhyGRU architecture to vehicle dynamics modeling. A comparison is carried out among three sequence-to-sequence predictors for vehicle dynamics: a vanilla GRU, a PhyGRU baseline and a PhyGRU with residual correction. The objective is to estimate the signals

$$\mathbf{y} = [a_x,\ a_y,\ r]^T \tag{1}$$

from the control input sequence

$$\mathbf{u} = [\delta,\ \text{throttle},\ \text{brake}]^T, \tag{2}$$

where a_x and a_y denote longitudinal and lateral accelerations, r denotes yaw rate and δ the steering input.

The three compared models are defined as follows. The GRU is a purely data-driven recurrent model. The PhyGRU baseline uses a physics-based bicycle prior inside the recurrent update and produces its output directly from the merging of physics model into the GRU gated structure. The PhyGRU residual uses the same physics-based recurrent update as the baseline, but its output is augmented by three learnable residual heads, one for each predicted channel.

2 Dataset generation and vehicle models

The datasets are generated synthetically by a nonlinear 7-DOF vehicle simulator and then used to train and evaluate the recurrent models. The simulator input is the control vector

$$\mathbf{u}_k = [\delta_k,\ \text{throttle}_k,\ \text{brake}_k]^T, \tag{3}$$

where δ_k is the steering angle, while throttle and brake are normalized actuation commands.

7-DOF ground-truth model. The ground-truth simulator state is

$$\mathbf{x}_k = [v_{x,k},\ v_{y,k},\ r_k,\ \omega_{fl,k},\ \omega_{fr,k},\ \omega_{rl,k},\ \omega_{rr,k}]^T, \tag{4}$$

where v_x and v_y are the longitudinal and lateral velocities, r is the yaw rate, and $\omega_{fl}, \omega_{fr}, \omega_{rl}, \omega_{rr}$ are the four wheel angular speeds.

The slip angles are computed as

$$\alpha_f = \delta_k - \arctan 2(v_{y,k} + l_f r_k,\ v_{x,k}^{\text{safe}}), \tag{5}$$

$$\alpha_r = -\arctan 2(v_{y,k} - l_r r_k,\ v_{x,k}^{\text{safe}}), \tag{6}$$

with a velocity regularization

$$v_{x,k}^{\text{safe}} = \max\left(|v_{x,k}|,\ v_{\min}\right) \operatorname{sign}(v_{x,k}). \tag{7}$$

The front and rear lateral tire forces are modeled as

$$F_{yf} = 2C_f \tanh(\alpha_f), \tag{8}$$
$$F_{yr} = 2C_r \tanh(\alpha_r). \tag{9}$$

The total longitudinal force is

$$F_x = F_{x,\max}\,\text{throttle}_k - F_{b,\max}\,\text{brake}_k - C_d v_{x,k}|v_{x,k}| - C_{rr}\tanh(v_{x,k}). \tag{10}$$

The planar rigid-body dynamics are

$$\dot{v}_{x,k} = \frac{F_x - F_{yf}\sin(\delta_k) + m v_{y,k} r_k}{m}, \tag{11}$$
$$\dot{v}_{y,k} = \frac{F_{yf}\cos(\delta_k) + F_{yr} - m v_{x,k} r_k}{m}, \tag{12}$$
$$\dot{r}_k = \frac{l_f F_{yf}\cos(\delta_k) - l_r F_{yr}}{I_z}. \tag{13}$$

The corresponding measurable outputs are

$$a_{x,k} = \dot{v}_{x,k} - v_{y,k} r_k, \qquad a_{y,k} = \dot{v}_{y,k} + v_{x,k} r_k, \qquad y_{3,k} = r_k. \tag{14}$$

Therefore, the output vector used in this work is

$$\mathbf{y}_k = [a_{x,k},\ a_{y,k},\ r_k]^T. \tag{15}$$

The wheel-speed dynamics are included only in the 7-DOF state evolution and are written as

$$\dot{\omega}_{fl,k} = \frac{-T_{brake,k} - R_w \frac{F_{x,f}}{2}}{I_w}, \tag{16}$$
$$\dot{\omega}_{fr,k} = \frac{-T_{brake,k} - R_w \frac{F_{x,f}}{2}}{I_w}, \tag{17}$$
$$\dot{\omega}_{rl,k} = \frac{T_{drive,k} - T_{brake,k} - R_w \frac{F_{x,r}}{2}}{I_w}, \tag{18}$$
$$\dot{\omega}_{rr,k} = \frac{T_{drive,k} - T_{brake,k} - R_w \frac{F_{x,r}}{2}}{I_w}, \tag{19}$$

where

$$T_{drive,k} = T_{d,\max}\,\text{throttle}_k, \qquad T_{brake,k} = T_{b,\max}\,\text{brake}_k, \tag{20}$$

and the longitudinal force is distributed between axles as

$$F_{x,f} = 0.4\,F_x, \qquad F_{x,r} = 0.6\,F_x. \tag{21}$$

Bicycle-model prior. A reduced bicycle model is used as the physics prior inside the PhyGRU. Its state is

$$\boldsymbol{\xi}_k = [v_{x,k},\ v_{y,k},\ r_k]^T, \tag{22}$$

and it uses the same input vector $\mathbf{u}_k$. The bicycle model retains the planar rigid-body equations and neglects the wheel-speed states. In the implementation, it uses a reduced parameter set $\boldsymbol{\theta}_{phy}$ and produces the same type of output vector,

$$\mathbf{y}_{phy,k} = [a_{x,k},\ a_{y,k},\ r_k]^T, \tag{23}$$

with

$$\alpha_f = \delta_k - \arctan 2(v_{y,k} + l_f r_k,\ v_{x,k}^{\text{safe}}), \tag{24}$$

$$\alpha_r = -\arctan 2(v_{y,k} - l_r r_k,\ v_{x,k}^{\text{safe}}), \tag{25}$$

$$F_{yf} = 2C_f \tanh(\alpha_f), \tag{26}$$

$$F_{yr} = 2C_r \tanh(\alpha_r), \tag{27}$$

$$F_x = F_{x,\max}\,\text{throttle}_k - F_{b,\max}\,\text{brake}_k - C_d v_{x,k}|v_{x,k}| - C_{rr}\tanh(v_{x,k}), \tag{28}$$

$$\dot{v}_{x,k} = \frac{F_x - F_{yf}\sin(\delta_k) + m v_{y,k} r_k}{m}, \tag{29}$$

$$\dot{v}_{y,k} = \frac{F_{yf}\cos(\delta_k) + F_{yr} - m v_{x,k} r_k}{m}, \tag{30}$$

$$\dot{r}_k = \frac{l_f F_{yf}\cos(\delta_k) - l_r F_{yr}}{I_z}, \tag{31}$$

$$a_{x,k} = \dot{v}_{x,k} - v_{y,k} r_k, \tag{32}$$

$$a_{y,k} = \dot{v}_{y,k} + v_{x,k} r_k. \tag{33}$$

Compared with the 7-DOF simulator, the bicycle prior is intentionally simpler because it does not include the wheel-spin states and is therefore a lower-order approximation of the same planar motion variables.

Data splitting and post-processing. Smooth random control sequences are generated by combining low-frequency sinusoidal components for steering, throttle, and brake. Three disjoint datasets are then simulated: 250 sequences for training, 50 sequences for validation, and 36 sequences for testing. The training and validation sequences have a duration of 10 s, while the test sequences have a duration of 50 s. This makes the test set substantially longer and therefore more demanding in terms of long-horizon consistency.

For each sequence, the simulator produces clean outputs $\mathbf{y}_k^{clean}$. Gaussian measurement noise is then added only to the outputs,

$$\mathbf{y}_k^{noisy} = \mathbf{y}_k^{clean} + \boldsymbol{\varepsilon}_k, \tag{34}$$

with zero-mean noise applied channel-wise. The noisy outputs are the quantities used for training and evaluation. Before learning, the output channels are standardized using the training-set statistics, while the inputs are kept in physical units. The same preprocessing is applied to validation and test data.

3 Learning setup

The baseline GRU is a vanilla recurrent model that directly maps the input sequence to the output sequence. Denoting the hidden state by $\mathbf{h}_k$, it can be written in compact form as

$$\hat{\mathbf{y}}_k = f_\theta(\mathbf{u}_{1:k}), \tag{35}$$

where the mapping is learned entirely from data.

The PhyGRU baseline uses the bicycle prior inside a gated recurrent update. Let the recurrent state be decomposed as

$$\mathbf{h}_k = \begin{bmatrix} \boldsymbol{\xi}_k \\ \boldsymbol{\ell}_k \end{bmatrix}, \tag{36}$$

where $\boldsymbol{\xi}_k \in \mathbb{R}^{n_\xi}$ is the physics prior state component and $\boldsymbol{\ell}_k \in \mathbb{R}^{n_\ell}$ is a latent state. The physics law produces a state derivative

$$\dot{\boldsymbol{\xi}}_{k-1} = f_{\text{phy}}(\boldsymbol{\xi}_{k-1}, \mathbf{u}_k; \boldsymbol{\theta}_{\text{phy}}), \tag{37}$$

which is integrated by a forward Euler step,

$$\boldsymbol{\xi}_k^{\text{cand}} = \boldsymbol{\xi}_{k-1} + \Delta t\, \dot{\boldsymbol{\xi}}_{k-1}. \tag{38}$$

The latent state is evolved by a learned linear dynamics branch,

$$\dot{\boldsymbol{\ell}}_{k-1} = W_{\text{lat}} \begin{bmatrix} \mathbf{h}_{k-1} \\ \mathbf{u}_k \end{bmatrix} + \mathbf{b}_{\text{lat}}, \tag{39}$$

and integrated as

$$\boldsymbol{\ell}_k^{\text{cand}} = \boldsymbol{\ell}_{k-1} + \Delta t\, \dot{\boldsymbol{\ell}}_{k-1}. \tag{40}$$

The candidate recurrent activation is then

$$\tilde{\mathbf{h}}_k = \begin{bmatrix} \boldsymbol{\xi}_k^{\text{cand}} \\ \boldsymbol{\ell}_k^{\text{cand}} \end{bmatrix}, \tag{41}$$

and the final state update is obtained through a GRU-style gate,

$$\mathbf{z}_k = \sigma\left(W_z \begin{bmatrix} \mathbf{h}_{k-1} \\ \mathbf{u}_k \end{bmatrix} + \mathbf{b}_z \right), \tag{42}$$

$$\mathbf{h}_k = (\mathbf{1} - \mathbf{z}_k) \odot \mathbf{h}_{k-1} + \mathbf{z}_k \odot \tilde{\mathbf{h}}_k. \tag{43}$$

In the baseline PhyGRU, the output is taken directly from the merging of bicycle model and recurrent gates structures, so no residual correction is added on the output.

The PhyGRU residual uses the same recurrent state update as the PhyGRU, but residual connections augments the output with a learned correction and exploit an MLP as latent. The correction term is produced by three independent small heads, one per output channel.

Before training, the outputs are standardized using the statistics of the training set, whereas the inputs remain in raw physical units. The same preprocessing is applied to the validation and test sets. All models are trained with mean squared error loss and the Adam optimizer. The training loop has been designed to bring the three models to comparable performance on validation, so as to better assess later the generalization on the test set with longer time series. Training was stopped once the models reached a comparable target validation MSE, ensuring a fair comparison of their generalization capabilities.

Model	Parameters	Epoch	Train MSE	Val MSE
GRU	13443	120	2.7886×10^{-2}	2.8131×10^{-2}
PhyGRU base	121	10	3.3438×10^{-2}	3.3380×10^{-2}
PhyGRU residual	1812	10	2.7187×10^{-2}	2.7396×10^{-2}

Table 1: Training and validation MSE together with model size (number of trainable parameters).

4 Evaluation protocol & Results

The test set is generated with the same synthetic procedure used for training and validation, so that the only difference lies in the held-out control sequences. Performance is evaluated on the noisy output signals through mean squared error and per-channel error over all testing trajectories.

The inference average time per-step are respectively of 0.0035 ms for GRU, 0.0270 ms for PhyGRU base, and 0.0229 ms for PhyGRU residual on Intel(R) Xeon(R) CPU @ 2.20 GHz.

Model	Total MSE	a_x	a_y	yaw rate
GRU	2.0510×10^{0}	4.8613×10^{-2}	6.1038×10^{0}	6.7830×10^{-4}
PhyGRU base	7.2265×10^{-2}	2.4674×10^{-2}	1.9195×10^{-1}	1.7366×10^{-4}
PhyGRU residual	4.2707×10^{-2}	2.3555×10^{-2}	1.0445×10^{-1}	1.1874×10^{-4}
Bicycle Prior	5.7738×10^{-2}	2.4300×10^{-3}	1.7068×10^{-1}	9.6260×10^{-5}

Table 2: Test MSE, total and per output component.

The base PhyGRU model, which uses a linear latent, was unable to deviate from the model provided in the prior; however, the residual PhyGRU version which has many more parameters and uses an MLP in the latent layer instead of a linear dynamic model improved upon the model provided in the prior, effectively blending model-based information with the data-driven component. These results in neural models are also affected by noise introduced during training, which the pure Bicycle model does not experience.

In addition, a qualitative trajectory reconstruction is performed by double integrating the predicted signals, in order to inspect the accumulated error in planar motion. The final result reported here is the grid of 36 reconstructed trajectories from the test set of 50 s duration. Each subplot shows the trajectory evolution for one test sequence, allowing a compact visual comparison of the predicted motion across the full test distribution.

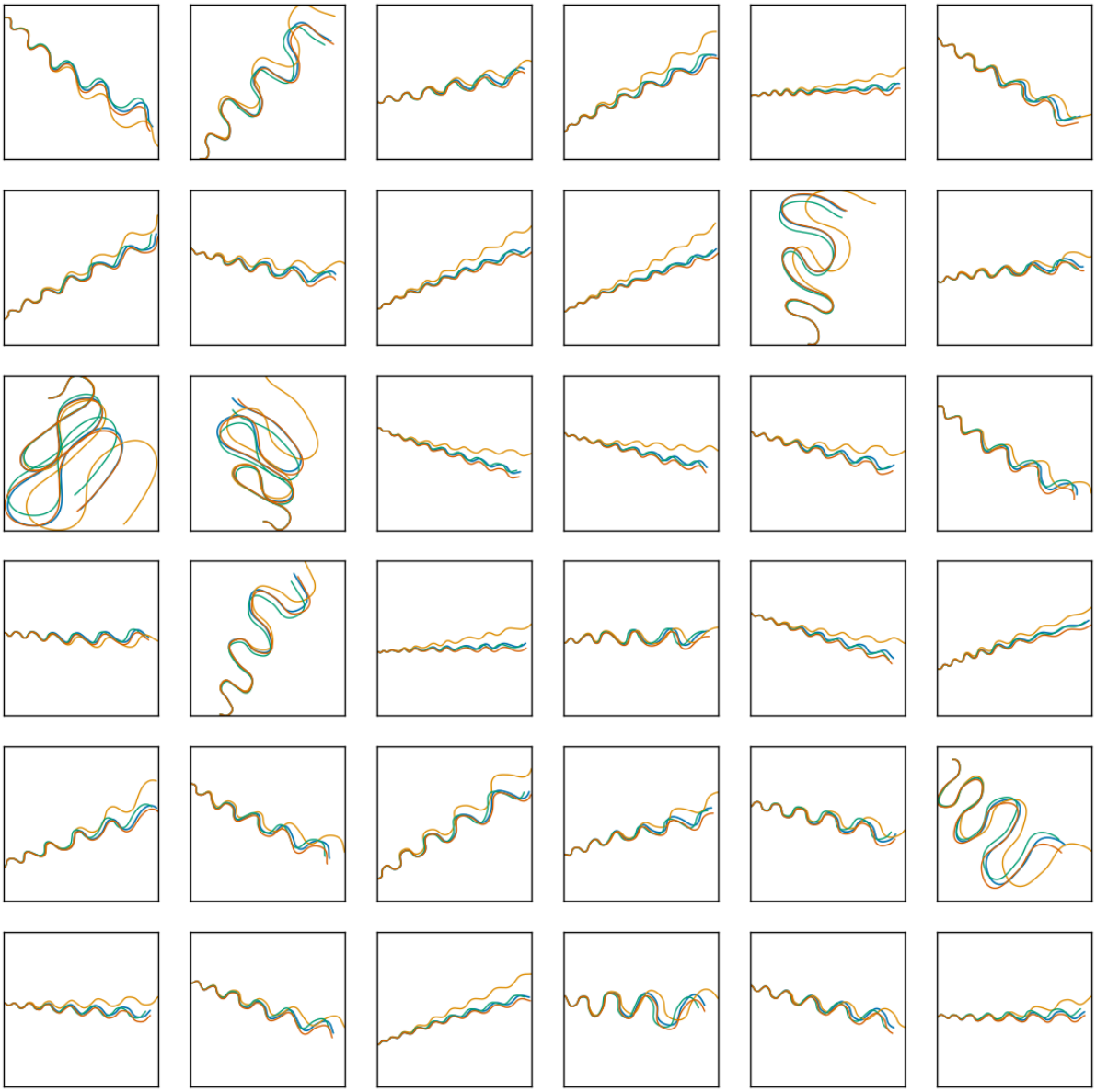

Figure 1: Test results, Ground Truth in blue, GRU in yellow, PhyGRU base green, and PhyGRU residual orange.

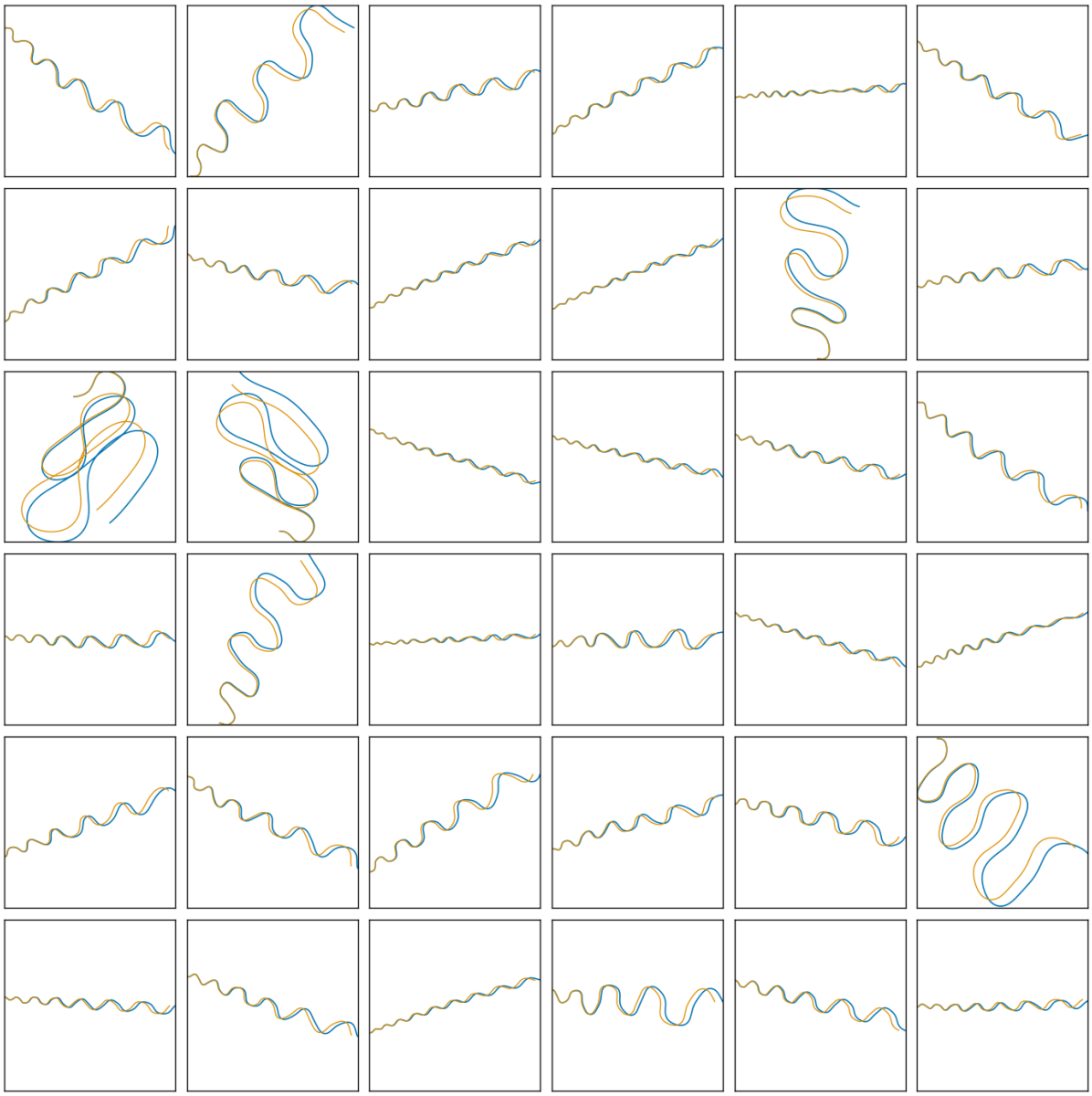

Figure 2: Test trajectories using 7-DOF model in blue and Bicycle model provided as prior in yellow.